

Aamira Bushra

+91 8007949684 | bushraaamira06@gmail.com | [LinkedIn](#) | [GitHub](#)

EDUCATION

Dayananda Sagar Academy of Technology and Management

Bengaluru, Karnataka

Bachelor of Engineering in Computer Science (4th Semester)

2024 – 2028

- **Relevant Coursework:** Data Structures, Analysis & Design of Algorithms, Operating Systems, DBMS, Object-Oriented Programming (Java), Computer Organization, Discrete Mathematics.

EXPERIENCE

Cloud Engineer & Project Lead

Jan 2026 – Present

Cyra AI - SOC Analysis Startup

Remote, India

- **Designed the MVP** for an **AI-powered security agent** for SOC analysis, defining product scope, technical architecture, and revenue model for investor pitches.
- Architected scalable, fault-tolerant cloud infrastructure on **GCP (GKE, Cloud Run, Cloud SQL, Vertex AI)** with centralized logging.
- Led migration of production workloads from **AWS to GCP** and transitioned managed services to open-source **NGINX**, improving scalability while optimizing operational cost.

PROJECTS

PurchaseAI | *Django REST, Next.js, TypeScript, ChromaDB, Ollama (Llama 3.2)*

Apr 2026 – Jun 2026

- Built a **full-stack procurement system** for my college (DSATM), exposing **18 REST endpoints** with **JWT** auth across a 9-role access model and a Next.js/TypeScript frontend.
- Engineered a local LLM assistant using **Ollama (Llama 3.2)** with a **RAG** pipeline over **ChromaDB**, using query expansion and distance-based relevance filtering.
- Added an LLM **tool-calling** layer routing queries to 6 Python analytics functions (trends, vendor ranking, anomaly detection), plus a CSV ingestion pipeline and web scrapers.

Zero-Copy Cryptographic Evaluation Framework for IoT | *Rust, Cryptography, QEMU*

Jan 2026 – May 2026

- Designed a **zero-copy** cryptographic model eliminating dynamic heap allocation to prevent fragmentation-induced OOM failures on IoT edge nodes with **<32KB** heap.
- Enforced **O(1)** buffers at compile time via **Rust's AeadInPlace** trait, benchmarking **AES-256-GCM, ChaCha20, and ASCON-128** across x86_64 and **QEMU**-emulated ARM64.
- Showed the naive model crashed (OOM) at iteration 212 while the zero-copy model ran **1,000 iterations at 0% failure**; documented in an IEEE-format paper being prepared for submission.

P2P Cloud Storage using Kademlia DHT | *Java, Distributed Systems, JUnit*

2025

- Implemented a decentralized P2P storage system on the **Kademlia DHT protocol** over a **160-bit XOR keyspace** for scalable, fault-tolerant data management.
- Built XOR-based routing, **k-buckets**, node lookup, and peer discovery, validated across **50 nodes** under churn with **JUnit** unit and integration tests.

PUBLICATIONS

Decentralized P2P Cloud Storage using the Kademlia DHT Protocol

2025

- Published in the *International Journal of Science and Engineering Technologies (IJSET)*; extended version submitted for peer review via **Microsoft CMT**.

TECHNICAL SKILLS

AI & LLM: RAG, Ollama (Llama 3.2), LLM Tool-Calling, Prompt Engineering, ChromaDB (Vector DB), Embeddings

Cloud & DevOps: GCP (GKE, Cloud Run, Vertex AI, IAM), AWS, Docker, Jenkins, Kafka, Terraform, NGINX

Cryptography: ChaCha20, ASCON, AES-GCM, ECC, ECDHE, SHA-256/3, RSA, OpenSSL

Backend & Systems: System Design, Django REST, Spring Boot, Node.js, FastAPI, Flask, PostgreSQL, Redis

Languages & Tools: C++, Java, Python, JavaScript, TypeScript, Rust, SQL, React.js, Next.js, JUnit, QEMU

Achievements: Solved **400+ LeetCode problems**; strong DSA and networking fundamentals.